

Nicky J. van den Berg

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PROFILE

I just finished my Ph.D. in (Applied) Mathematics, specializing in geometric image processing. My motivation is fueled by the feeling that my research can have a positive impact on our society. Therefore, my personal goal is to develop image processing techniques that improve the quality of life.

Previously, I have focused on medical applications. I developed techniques to process retinal images, such as tracking and automatic grouping of vascular trees. I am eager to contribute to the development of better image processing techniques for applications that can change the world.

EXPERIENCE

PhD Candidate, Eindhoven University of Technology (TU/e) 11 - 2021 – 11 - 2025

Title Thesis: *Geometric Tracking and Grouping of Complex Vascularities in Medical Images*.

Supervision: Remco Duits, Olga Mula, Tos Berendschot

Financed by the NWO Talent Programme VICI 2020 Exact Sciences (Duits, Geometric Learning for Image Analysis, VI.C 202–031).

Visiting Researcher, Emory 10 - 2024 – 10 - 2024

Host: Lars Ruthotto

Financed by the EU REMODEL program (MSCA-SE 101131557).

Teaching Assistant, TU/e 09 - 2015 – 08 - 2021

Supervising tutorials for students from various programs (group size between 8 and 120 students) and correcting homework exercises.

Technical Author, Numworx 12 - 2019 – 06 - 2020

Developing mathematics exercises at a high school and college level for a digital mathematics platform.

Student Mentor, TU/e 09 - 2015 – 02 - 2017

Guiding freshmen in the transition to university.

EDUCATION

Eindhoven University of Technology (TU/e) 04 - 2018 – 07 - 2021

Master Industrial and Applied Mathematics cum laude

Master Thesis: *Improved Computational Schemes for Geodesic Vessel Tracking in Orientation Scores*

Supervision: Remco Duits, Bart M.N. Smets

University of Western Ontario (UWO) 02 - 2020 – 06 - 2020

Research Internship

Title Research Report: *The Blood Coagulation Process: A Mathematical Model*

Supervision: Mikko Karttunen, Barry Koren

Eindhoven University of Technology (TU/e) 09 - 2014 – 03 - 2018

Bachelor Applied Mathematics with courses from Psychology & Technology

Bachelor Thesis: *Disease Dynamics in the Framework of Interacting Particle Systems: The Position Dependent SIS-model*

Supervision: Oliver Tse

Provinciale Secundaire School Voeren (PSS Voeren); High School 09 - 2008 – 06 - 2014

Modern Languages - Science

Extra-curricular courses in mathematics and science (physics and chemistry)

EXTRA-CURRICULAR ACTIVITIES

Founder and Board Member, Stichting Ada Alumni *04 - 2022 – 01 - 2026*
Founding a foundation for the Mathematics and Computer Science department alumni at TU/e and organizing activities for this audience.
Functions: Secretary (*04 - 2022 – 10 - 2022*), Treasurer (*10 - 2023 – 01 - 2026*).

Committee Member, E.P.A. Nexus (PhD association) *04 - 2023 – 11 - 2024*
Representation Committee (*04 - 2023 – 11 - 2024*) & Cash Control Committee (*11 - 2023 – 11 - 2024*)

Student Member Program Committee Mathematics, TU/e *09 - 2015 – 04 - 2021*
Checking the quality of education (in co-determination). Representing the program committee in various general meetings, accreditations, and the ITK (quality assurance institutional assessment, see website NVAO).

Student Member Department Council, TU/e *01 - 2018 – 12 - 2019*
Advising the department board while focusing on communication, well-being, and education.

Secretary, Study association GEWIS *07 - 2017 – 06 - 2018*
Responsible for the well-being of the association during the academic year 2017-2018.
Responsible for the member administration, minutes of board and general meetings, and all mail.

Various Committees, Study association GEWIS *09 - 2014 – 10 - 2021*
Including developing a multi-annual vision, corporate identity, revision of the bylaws and internal regulations, etc.

Activities in High School, PSS Voeren *09 - 2012 – 06 - 2014*
Chair and member of the student council.
Member of the school council.

OTHER ACTIVITIES

Workshop for high school teachers on AI, 4TU.NIRICT *11 - 2025 – 11 - 2025*
Presenting current image processing research (partially relying on) AI to mathematics high school teachers. Financed by 4TU.NIRICT to raise awareness in high schools on ICT challenges.

Organization of CASA outing, TU/e *05 - 2025 – 06 - 2025*
Organizing a day for the cluster consisting of a small activity, two museum visits, and lunch and dinner.

Lil'Scientist *03 - 2024 – 06 - 2024*
Guiding primary school students through their first research project. This is a project of “De Jonge Akademie”.

PROOF - PhD Development Sounding Board, TU/e *09 - 2023 – 09 - 2023*
Improving the experience and offer of PROOF-courses for PhD candidates.

Panel Discussion Research Day, TU/e *06 - 2023 – 06 - 2023*
On the impact of research.

Organization of CASA day, TU/e *06 - 2022 – 05 - 2023*
Organization of a day for the cluster where ongoing research is discussed, combined with a skill-session (e.g. workshop presenting skills, or a relevant lecture).

Curriculum Committee, TU/e *02 - 2022 – 07 - 2022*
Giving feedback on the newly developed curriculum for the bachelor Applied Mathematics.

Research Assessment, TU/e *03 - 2022 – 03 - 2022*
Part of the PhD candidate representation during interviews with the panel.

TEACHING

Tutor/teaching assistant for the following courses:

Numerical Linear Algebra Applied Mathematics students	2024 – 2024
Complex Analysis Applied Mathematics students	2022 – 2024
Calculus Built Environment students	2022 – 2024
Mathematics 1 Electrical Engineering and Industrial Engineering students	2018 – 2021
Calculus Applied Physics students	2017 – 2019
Data Analytics for Engineers Electrical Engineering, Applied Mathematics, and Psychology & Technology students	2017 – 2018
Introduction to Modeling: From Problems to Numbers and Back Psychology & Technology students	2016 – 2017
Calculus Mechanical Engineering, and Medical Sciences & Technology students	2015 – 2017

SUPERVISION

Bachelor Project , Franciska Asma Title Thesis: <i>Curvature Estimation and Adaptation for Chip Components Segmentation in SEM Images</i>	02 - 2024 – 11 - 2024
Graduation Project , Leanne Vis Title Thesis: <i>Automatic Edge-detection and Grouping of Possibly Overlapping Components in SEM Images of Silicon Wafers</i>	09 - 2023 – 07 - 2024
Modeling Week Project Title Report: <i>Geodesic Tracking of Blood Vessels in Wide-Field Retinal Images</i> (in $SO(3)$)	11 - 2023 – 11 - 2023
Research Topic , Leanne Vis Title Research Report: <i>Connected Components</i>	03 - 2023 – 06 - 2023
Research Internship , Romain Dugast Title Research Report: <i>Geometric Tracking in $SE(2)$ and PDE-G-CNNs for the Classification of Arteries and Veins</i>	04 - 2022 – 08 - 2022

PRESENTATIONS

Real World Applications of Geometry and Algebra (RAGA) 2025 , Eindhoven University of Technology Title: <i>Geodesic Tracking of Vascular Trees in Retinal Images</i>	27 - 05 - 2025
CASA Day , Eindhoven University of Technology Title: <i>Geodesic Tracking on Retinal Images</i>	16 - 04 - 2025
Ophthalmology Seminar , Maastricht University Title: <i>Image Analysis on Retinal Images</i>	14 - 03 - 2025
Algebra Seminar , Georgia Tech Title: <i>Lie Groups and Applications to Multi-Orientation Image Analysis</i>	07 - 10 - 2024

Research Talk, Emory 01 - 10 - 2024
 Title: *Lie Groups and Applications to Multi-Orientation Image Analysis*

Geometry and Machine Learning, Conference 19 - 06 - 2024
 Title: *Geodesic Tracking via Data-Driven Geometry for Vascular Tree Tracking*

Ophthalmology Seminar, Maastricht University 31 - 03 - 2023
 Title: *Geodesic Tracking of Retinal Vascular Trees with Optical and TV-Flow Enhancement in the Lifted Space of Positions and Orientations $SE(2)$*

CASA Day, Evoluon 13 - 04 - 2022
 Title: *Geodesic Tracking via New Data-Driven Cartan Connections for Vascular Tree Tracking*

POSTERS

Crossing-Preserving Geodesic Tracking on Spherical Images, SSVM 2025 05 - 2025
Geodesic Tracking of Retinal Vascular Trees with Optical and TV-Flow Enhancement in $SE(2)$, NMC 2024 04 - 2024
Geodesic Tracking of Retinal Vascular Trees with Optical and TV-Flow Enhancement in $SE(2)$, SSVM 2023, CASA day 2024 05 - 2023, 11 - 2023

PUBLICATIONS

van den Berg, N.J., Vis, L., Mula, O., Duits, R. (in revision). Connected Components on Lie Groups and Applications to Multi-Orientation Image Analysis. <https://arxiv.org/abs/2409.18002>

García-Castellanos, A., Wessels, D.R., **van den Berg, N.J.**, Duits, R., Pelt, D.M., Bekkers, E.J. (accepted to NeurIPS). Equivariant Eikonal Neural Networks: Grid-Free, Scalable Travel-Time Prediction on Homogeneous Spaces. <https://arxiv.org/abs/2505.16035>

van den Berg, N.J., Sherry, F.M., Berendschot, T.T.J.M., Duits, R. (2025). Crossing-Preserving Geodesic Tracking on Spherical Images. In: Bubba, T.A., Gaburro, R., Gazzola, S., Papafitsoros, K., Pereyra, M., Schönlieb, CB. (eds) Scale Space and Variational Methods in Computer Vision. SSVM 2025. Lecture Notes in Computer Science, vol 15668. Springer, Cham. https://doi.org/10.1007/978-3-031-92369-2_15

van den Berg, N.J., Smets, B.M.N., Pai, G., Mirebeau, J.-M., Duits, R. Geodesic Tracking via New Data-Driven Connections of Cartan Type for Vascular Tree Tracking. J Math Imaging Vis (2024). <https://doi.org/10.1007/s10851-023-01170-x>

van den Berg, N.J., Zhang, S., Smets, B.M.N., Berendschot, T.T.J.M., Duits, R. (2023). Geodesic Tracking of Retinal Vascular Trees with Optical and TV-Flow Enhancement in $SE(2)$. In: Calatroni, L., Donatelli, M., Morigi, S., Prato, M., Santacesaria, M. (eds) Scale Space and Variational Methods in Computer Vision. SSVM 2023. Lecture Notes in Computer Science, vol 14009. Springer, Cham. https://doi.org/10.1007/978-3-031-31975-4_40

Kompanets, A., Duits, R., Leonetti, D., **van den Berg, N.**, Snijder, H.H. (2024). Segmentation Tool for Images of Cracks. In: Skatulla, S., Beushausen, H. (eds) Advances in Information Technology in Civil and Building Engineering. ICCCB 2022. Lecture Notes in Civil Engineering, vol 357. Springer, Cham. https://doi.org/10.1007/978-3-031-35399-4_8

SKILLS

Programming Languages and Frameworks Python (incl. PyTorch), Mathematica, MATLAB, $\text{\LaTeX}$
Languages Dutch (native), English (fluent), French (intermediate), German (intermediate)